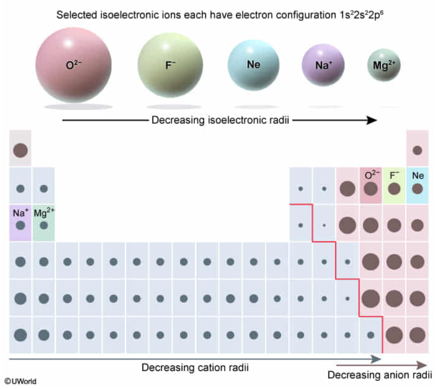
The regulation of mineral ions in the cellular fluids of biological systems is performed by ion pumps that selectively transport ions of a specific size (ionic radius) and charge across cellular membranes. Which of the following isoelectronic species is the smallest ion?

- A. Na^+ [20%]
- B. F^- [32%]
- ✔ C. Mg^{2+} [35%]
- D. O^{2-} [11%]

Correct
34% answered correctly

Explanation:



The radius of an atom or ion is the distance from the center of the nucleus to the outermost electron shell. The atomic radius is the measurement for neutral atoms that retain all their electrons whereas the ionic radius is the measurement for atoms that have gained or lost electrons (often an entire electron shell). As such, trends in these two measurements are distinct and must not be treated equivalently.

Ionic radii tend to **decrease in size across a period** (row) of the periodic table (left to right) and **increase moving down a group** (column). This trend **occurs for metal cations**, and then resets and repeats for **anions** beginning near the division between metals and nonmetals, past which anions tend to preferentially form. The size of an ion does not refer to its mass: Two ions may have the same spherical size (radius) from the nucleus to the edge of the electron cloud but have different masses depending on the number of protons and neutrons in the nucleus.

Compared to the **neutral atom** of a given element, its cation will be smaller but its anion will be larger. Losing electrons to form a cation causes the remaining electrons to experience a greater effective nuclear charge (Z_{eff}), pulling the electrons closer to the nucleus. Conversely, gaining electrons to form an anion produces greater electronic repulsion and nuclear shielding (lesser Z_{eff}), which pushes electrons farther from the nucleus.

Na^+, F^-, Mg^{2+}, and O^{2-} ions are **isoelectronic** (have the same number of electrons) but because the number of protons is different in each ion, the electrons in each ion experience a different Z_{eff}. Therefore, in an isoelectronic series, ionic radii decrease with increasing atomic number. Because Mg^{2+} has the highest atomic number (greatest number of protons) in this isoelectronic series, Z_{eff} is greatest in this ion, making it the smallest within the given series.

(Choice A) Cations are generally smaller than neutral atoms and anions in an isoelectronic series, but Na^+ is not the smallest because it has one less proton than Mg^{2+} (ie, lesser Z_{eff} to pull the electrons toward the nucleus).

(Choices B and D) Anions in an isoelectronic series are generally larger than neutral atoms and cations in the same series.

Educational objective:
Ionic radii decrease in size across a period and increase down a group on the periodic table. The use of isoelectronic series is helpful in comparing cations, anions, and neutral atoms.

Time spent: 0 sec
Subject: General Chemistry

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